Biochemistry Cume
September 12, 2008

1. Explain how the noncovalent binding of a molecule, such as a ligand like O_2, to a protein might change the protein's conformation. Also discuss if the primary structure of the protein be changed by such noncovalent binding? (25 pts.)

2. In an aqueous environment, will the following peptide fragment more likely be buried inside a globular protein or located on its surface? Explain your answer. (30 pts.)

Leu-Ala-Gly-Pro-Phe-Ser-Val-Asp-Ile-Phe

3. A solution of a peptide of unknown sequence was divided into 2 samples. One sample was treated with trypsin, and the other sample treated with chymotrypsin. The peptide sequences detected after trypsin treatment had the following sequences:
A-A-W-G-K

T-F-V-K

The smaller peptides obtained after the chymotrypsin treatment had the following sequences:
V-K-A-A-W
G-K
T-F

What is the sequence of the original peptide, and describe what trypsin and chymotrypsin do in each of their respective treatments. (15 pts.)

4. Membrane-bound proteins are generally closely associated with nonpolar regions of lipid molecules. Speculate as to how detergents might affect the structural integrity of membrane-bond proteins and soluble proteins. (30 pts.)

B